

Kenza Tazi

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EDUCATION	<u>University of Cambridge</u>, MRes + PhD Environmental Data Science 2019-2025 <i>PhD thesis: Precipitation prediction over High Mountain Asia using Gaussian processes</i> • Focused on improving historical and future precipitation estimates, modelling the probability of extremes events that lead to floods and droughts for the 1.9 billion people who rely on these mountains for water security. • Applied wide range of deep learning models including convolutional neural networks and neural processes and developed novel Gaussian process models, including climate model emulators. • Used a large range of hydrological datasets, including weather station data, satellite observations, reanalysis and climate model outputs. • Supervised by Richard Turner (Uni. of Cambridge), Scott Hosking (Alan Turing Institute), Andrew Orr (British Antarctic Survey) and Javier Gonzalez (Microsoft). <i>MRes thesis: Precipitation prediction in the Upper Indus Basin using Gaussian processes</i> Probabilistic machine learning model for future precipitation predictions in key area for Pakistani, Indian, and Chinese water security. <u>Imperial College London</u>, MSci Physics 2015-2019 <i>MSc thesis: Cloud identification in satellite images using artificial intelligence</i> Deep learning model to improve cloud identification for Sentinel 3 satellites by over 30% over polar regions where clouds are most important to global radiation forcing and most challenging to identify. <i>BSc thesis: Modelling the behaviour, occurrence and emissions of wildfire on a global scale</i>
EXPERIENCE	<u>European Centre for Medium-Range Weather Forecasts (ECMWF)</u> 2025 • Operationalised centre's first machine learning global flood model, AIFL. • Leading development of a probabilistic global flood forecasting model. • Researching various aspects of model improvement including feature selection, data processing, model architecture, and code optimisation. • Integrating hydrology and land surface processes into the AIFS, the centre's machine learning weather model. • Managing EUR 300,000 contract focused on an agentic LLM framework for hydrology and water quality. <u>Frontier Development Lab</u> 2022 • Led team of PhD students and post-doctoral researchers to study pyroCb clouds associated with the most intense and unpredictable wildfires with support from Google, NVIDIA, and the European Space Agency. • Created the first global pyroCb database and machine learning forecasting system and conducted causal invariance modelling to better understand pyroCb drivers. <u>Geophysical Fluid Dynamics Group, University of Oxford</u> 2018 • Investigated the 2016 stalling of the Quasi-Biennial Oscillation through laboratory experiments. • Designed framework to simultaneously run twelve motors in different wave patterns to generate pseudo-gravity waves in a water filled annulus and analysed footage of the waves using particle image velocimetry.

Planetary Science Group, University of Oxford 2017

- Designed and built a light source for evaluating three-dimensional thermal emissions from lunar and asteroid samples with a cooling system.
- Built electronic interface to move two-axis platform and measured performance of light source and radiometer.

British Antarctic Survey, Cambridge 2016

- Wrote program to calculate diurnal and nocturnal cloud cover for wide field cameras using model simulations, image analysis, multi-coordinate system mapping.
- Software deployed at the Halley and Rothera bases in Antarctica and used for microphysical cloud property and gravity wave research.

AWARDS	Best student precipitation presentation award, AGU23	2023
	First place in Cambridge ICCS Reproducibility Hackathon	2022
	AJ Pressland Fund, University of Cambridge	2022
	AI for Earth Grant, Microsoft (USD25,000)	2021
	AI for Environmental Risk (AI4ER) CDT Studentship	2019
	Stevenson Award, Imperial College London	2019
	Student Award for Outstanding Achievement, Imperial College London	2019
	Dean's Fund, Imperial College London	2018
	Royal Astronomical Society Grant	2018

PUBLICATIONS * presented at international conference

Taccari M, **Tazi K**, Morrison OM, Grafberger A, Colonese J, Carton de Wiart C, Prudhomme C, Mazzetti C, Chantry M, Pappenberger F (2026). AIFL: A Global Daily Streamflow Forecasting Model Using Deterministic LSTM Pre-trained on ERA5-Land and Fine-tuned on IFS. *Journal of Hydrology*. *

Tazi K, Kim SWP, Girona-Mata M, Turner RE (2025). Refined climatologies of future precipitation over High Mountain Asia using probabilistic ensemble learning. *Environmental Research Letters*. *

Tazi K, Orr A, Hosking JS, Turner RE (2025). Precipitation prediction over the Upper Indus Basin from large-scale circulation patterns using Gaussian Processes. *Environmental Data Science*. *

Tazi K, Orr A, Hernandez-González J, Hosking JS, Turner RE (2024). Downscaling precipitation over High Mountain Asia using multi-fidelity Gaussian Processes: Improved estimates from ERA5. *Hydrology and Earth System Science*. *

Tazi K, Lin JA, Viljoen R, Gardner A, John T, Ge H, Turner RE (2023). Beyond intuition, a framework for applying Gaussian processes to real-world data. In *ICML 2023 Workshop on Structured Probabilistic Inference and Generative Modelling*. *

Tazi K, Salas-Porras ED, Braude A, Okoh D, Lamb KD, Watson-Parris D, Harder P, Meinert N (2022). Pyrocast: A machine learning pipeline to forecast pyrocumulonimbus (pyroCb) clouds. In *NeurIPS 2022 Workshop Tackling Climate Change with Machine Learning*. *

Salas-Porras ED, **Tazi K**, Braude A, Okoh D, Lamb KD, Watson-Parris D, Harder P, Meinert N (2022). Identifying causes of pyrocumulonimbus (pyroCb). In *NeurIPS 2022 Workshop on Causality for Real-world Impact*. *

Lalchand V, **Tazi K**, Cheema TM, Turner RE, Hosking JS (2022). Kernel Learning for Explainable Climate Science. In *UAI 2022 Workshop on Bayesian Modelling Applications*. *

Poulsen C, Egede U, Robbins D, Sandeford B, **Tazi K**, & Zhu T. (2020). Evaluation and comparison of a machine learning cloud identification algorithm for the SLSTR in polar regions. *Remote Sensing of Environment*.

TEACHING

Lectures

- Machine learning for weather and flood forecasting, WIRED Summer Lab (2025)
- Gaussian processes in practice, NERC Bayesian Machine Learning as a Tool for Climate Scientist Workshop (2024)
- FAIR data practices, AI4ER CDT (2020, 2021, 2022, 2023)

Workshops

- Weekly pair programming sessions for AI4ER CDT students (2021-2025)
- ECMWF ML Journal Club co-chair (2026-2027)

Supervision

- Small group problem sheet supervisions for the Department of Engineering's Inference course (3F8, 2025)
- Example classes for AI4ER CDT students auditing Advanced Machine Learning (MLMI4, 2022) and Machine Learning & Bayesian Inference (Part II, 2023)
- Supervision and assessment of students undertaking their 3rd year projects for the Department of Engineering. Topics ranged from civil, mechanical, information and bioengineering (2021, 2022, 2023)

Tutoring

- Private tutoring with a focus on maths, computer science and physics at high school and undergraduate level (2018-2022)

INVITED TALKS

Advances in hydrological variables in the AIFS

CaMa-Flood developer/user international meeting, *ECMWF* Sep 2026

AIFL: Towards a data-driven operational flood forecasting model

Royal Meteorological Society/University of Reading, *Public Lecture Series* May 2026

University of Reading, *Hydrology ML Seminar* Oct 2025

University of Oxford, *GRIT & Global River Bifurcations Workshop* Sep 2025

Changing the paradigm in operational weather and flood forecasting

WIRED Summer Lab Jul 2025

Precipitation prediction from over the Upper Indus Basin using Gaussian processes

The Alan Turing Institute, *Environment & Sustainability Seminar Series* Oct 2024

Ensembling climate models with Gaussian processes to better predict future extremes

University of Leeds, *SciML Seminar Series* Oct 2024

University of Cambridge, *Atmospheric Chemistry Group* Oct 2024

AGU, *Precipitation Technical Committee Seminar* Mar 2024

Improving precipitation predictions over High Mountain Asia using Gaussian processes

University College London, *Environment and Sustainability Group* Jul 2024

Narrowing precipitation uncertainty over High Mountain Asia using probabilistic ML

Shanghai AI Lab, *Lu Group* May 2024

MILA – Québec AI Institute, *Rolnick Group* Apr 2024

NERC Bayesian Machine Learning for Climate Scientists Workshop Mar 2024

NASA Jet Propulsion Laboratory, *SUDS Seminar* Dec 2023

Morocco AI, *Research Webinar Series* Jul 2023

University of Cambridge, *Energy and Environment Group* Jun 2023

Gaussian processes in practice

	NERC Bayesian Machine Learning for Climate Scientists Workshop	Mar 2024
	<u>Pyrocast: A machine learning pipeline to forecast pyrocumulonimbus clouds</u> University of Cambridge, <i>AI for Environmental Risk CDT</i>	Nov 2022
CONFERENCES	Climate Informatics (talk)	2024
	AGU Fall Meeting (poster and talk)	2023
	ICML – Probabilistic Inference & Generative Modelling Workshop (poster)	2023
	AI for Environmental Risk CDT showcase (talk)	2023
	NeurIPS– Tackling Climate Change with AI Workshop (poster)	2022
	Climate Informatics (poster)	2022
	Lunar and Planetary Science Conference (poster)	2018
ACADEMIC SERVICE	<u>Programme leadership</u> High Mountain Data Co-Lead for the Himalayan University Consortium (2023-2026)	
	<u>Conferences and workshops</u> • ECMWF ‘ML for Land and Hydrology Workshop’ (2026) – three-day event to bring together international leaders and scientists and discuss state-of-the-art and operational methods • University of Cambridge ‘Stochastic Processes Workshops’ (2021, 2023) – one-day events to collaborate on applications to real-world problems	
	<u>Outreach</u> • BCG ‘Climate and Sustainability Stewardship’ Programme (2022-2024) • Murray Edwards College ‘She Talks Science Webinar’ (2021, 2023) • Raspberry Pi magazine: Hello World (2022), Issue 19: Sustainability & Computing • Rocket Seeds - see Fernando B, Wade J, Tazi K. (2016) Sowing seeds from space. Astronomy & Geophysics. 2016 Oct 1;57(5):5-11	
	<u>Reviews:</u> Climate Informatics; ICML – Structured Probabilistic Inference and Generative Modelling Workshop; Journal of Geophysical Research - Atmospheres	
	<u>Committees</u> • AI4ER CDT Management Committee (2021-2023) • AI4ER CDT Ethics Diversity and Inclusion Committee (2022-2023) • Undergraduate Student Staff Committee, Department of Physics, Imperial College London (2016-2017) • University Student Network Executive Committee, Institute of Physics (2016-2017)	
POLICY	<u>Polar Summit, Paris Peace Forum</u> , invited delegate	2023
	Worked collectively with other scientists to secure EUR 1 billion pledge towards polar and high mountain research from French government.	
	<u>Cambridgeshire County Council</u> , consultant	2020-2021
	Undertook research for the Council through the Cambridge University Science and Policy Exchange (CUSPE) creating the Cambridgeshire Decarbonisation Fund, a new policy framework to decarbonise the county by 2050.	
	<u>All Party Parliamentary Group on Air Pollution</u> , lead author	2020
	Guided a small team to submit evidence on ways to keep low air pollution levels as UK exited the first Coronavirus Lockdown. Measures including making temporary cycle and pedestrian lanes were implemented.	
	<u>Tsinghua University’s Environment Summer School</u> , invited delegate	2017

Imperial delegate sent to design and pitch policy project to overcome one of China's environmental challenges to leading academics and policymakers.

OTHER EXPERIENCE	<u>Clean Planet Peninsula</u>	2024-2025
	Studied and conducted microplastic sampling in extreme environments including Dartmoor National Park, UK and Ymerbukta Glacier, Svalbard	
	<u>EntrepriseTech</u> , University of Cambridge	2020-2021
	Led team to propose business plan for a drug-discovery start-up with mentorship from the Head of Strategy to the VP of Artificial Intelligence at AstraZeneca Cambridge.	
LANGUAGES	<u>Science Museum</u> , London	2018-2019
	Advised curators on instrument displays and public engagement for the ‘London: City of Science’ permanent exhibit.	
	<u>Winter Olympic Games</u> , Sochi	2014
OTHER SKILLS	Represented Morocco in the Women’s Alpine Skiing Giant Slalom and Slalom events and competed in international circuits (FIS races, South America Cup and French Cup).	
	English and French (fluent), Korean (TOPIK Level 3), German (prev. C1) Python (incl. TensorFlow, PyTorch, Jax), Julia, MATLAB, Arduino, R, HTML, CSS	
OTHER SKILLS	Cloud and high-performance computing, manufacturing and graphic design training	